Therapeutic Potential of Complement Receptor Blockade in Sjögren's Syndrome and Lupus Erythematosus

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Sjögren's Syndrome (SS) and Systemic Lupus Erythematosus (SLE) are chronic systemic autoimmune diseases; aberrant immune activation, autoantibody production, and end-organ inflammation are common characteristics. SS affects approximately 0.1–0.4% of the general population; SLE affects 20–150 per 100,000 individuals globally, both with a striking 9:1 female predominance and significant systemic disease burden. Current therapeutic options remain inadequate; no FDA-approved disease-modifying therapies exist for SS, and while advances have been made in SLE management, a substantial proportion of patients continue to experience disease flares and progressive organ damage. Critically, no approved therapy directly targets the complement cascade in either disease, representing a significant therapeutic gap. Dysregulation of the complement cascade has long been implicated as a central pathogenic driver in both SS and SLE, contributing to immune complex accumulation, type I interferon signaling, and progressive tissue injury. Aberrant activation of terminal complement proteins drives the pathogenesis of SS and SLE through dysregulation of the alternative pathway, a constitutively active cascade that serves as a critical amplification loop for complement activation. Perturbations of the alternative pathway enable the overproduction of proinflammatory mediators by impairing regulatory control over the central complement proteins C3 and C5. Furthermore, differential expression of complement receptors CR1, CR2, CR3, and CR4 in SS and SLE perpetuates this proinflammatory milieu, amplifying immune-mediated tissue injury.

Complement System | Sjögrens Syndrome | Lupus Erythematosus | Therapeutics
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Introduction

Sjögren's Syndrome (SS) and Systemic Lupus Erythematosus (SLE) are chronic systemic autoimmune diseases that collectively affect millions of individuals worldwide; both are observed disproportionately in women and minority populations. The global incidence rate of SS is approximately 7 per 100,000 person-years (1), with higher incidence and prevalence rates observed in women and older age groups (2); onset typically occurs in the fourth and fifth decade of life. Beyond the distinctive sicca symptoms, SS carries significant systemic burden; a 2021 meta-analysis demonstrated a nearly 50% increase in mortality among patients with primary Sjögren's syndrome (pSS), with older age, male gender, vasculitis, hypocomplementaemia, positive anti-La/SSB, and cryoglobulinaemia identified as adverse prognostic factors (3). The global incidence rate of SLE is approximately 5.14 per 100,000 person-years, with the rate differing markedly between sexes; for men the rate is 1.53 per 100,000 person-years while for women this rate increases to 8.82 per 100,000 person-years (4). SLE predominantly affects young women, with Black, Hispanic, and Asian populations experiencing increased prevalence, incidence, and morbidity (5). Lupus nephritis (LN) is the most common cause of kidney injury in those afflicted with SLE, occurring in approximately 50% of patients; male patients tend to have more aggressive disease with higher rates of renal and cardiovascular involvement (6). There is an established clinical and serological overlap between SS and SLE; co-occurrence rates are approximately 23%, with the SLE-SS group being older and more female (7); both diseases share similar anti-SSA and anti-SSB autoantibody profiles, hypergammaglobulinemia, and complement consumption (8). Despite their prevalence and significant morbidity, current therapeutic options for both diseases remain inadequate. No effective disease-modifying therapy has been approved for SS, with current treatment largely focused on symptom management; clinical trials on targeted drugs have frequently failed due to an inability to demonstrate efficacy from misidentified targets or inconsistent research outcomes. For SLE, advancements have been made in disease management through therapeutics such as belimumab and anifrolumab; however, the complexity of clinical diagnosis and disease management remains a major challenge, with an over-reliance on corticosteroid therapy contributing to long-term organ damage and substantial morbidity. Critically, no approved therapy directly targets the complement cascade in either SS or SLE, representing a significant and unmet therapeutic need.

The complement system is a critical arm of innate immunity comprising over 50 soluble and membrane-bound proteins that function in a tightly regulated cascade. Activation proceeds through three distinct pathways — the classical, lectin, and alternative pathways — all converging at the cleavage of the central complement protein C3 into C3a, a potent anaphylatoxin, and C3b, a major opsonin, which subsequently drives cleavage of C5 into C5a and C5b, initiating assembly of the membrane attack complex (MAC). The complement activation via the alternative pathway has been recognized as

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a major pathogenic mechanism in autoimmune conditions; the alternative pathway is constitutively active and serves as a critical amplification loop for complement activation regardless of the initiating pathway (9). Tight regulation of this amplification loop by factor H, factor I, and membrane-bound regulatory proteins is essential to prevent self-tissue damage; perturbation of these regulators permits unchecked C3 and C5 activation with consequent inflammatory tissue injury.

Dysregulation of the complement cascade has long been implicated as a central pathogenic driver in both SS and SLE. A genetic pattern involving partial deficiencies of C2 and C4A in the classical complement pathway has been demonstrated to be a strong risk factor for both SLE and primary SS, emphasizing the central role of the complement system in the pathogenesis of both diseases. PubMed In SLE, the pathogenic role of complement is paradoxical — inherited deficiencies of classical pathway components impair the tolerogenic clearance of apoptotic debris, yet complement simultaneously drives inflammatory tissue injury when aberrantly activated by the immune complexes that accumulate as a consequence of this clearance failure. Patients with SLE have relative deficiencies of the C3b/C4b receptor CR1 (CD35) on erythrocytes; this receptor mediates the binding, transport, and endocytosis of circulating immune complexes, and its deficiency is fundamental to the pathogenesis of SLE. PubMed In SS, the characteristic anti-SSA/Ro52 autoantibodies suggest complement activation via the classical pathway through immune complex formation and tissue damage; moreover, a decrease in serum C3 and C4 levels is detected in over 20% of SS patients, indicating the role of innate immunity in SS pathogenesis. PubMed Central

Central to the complement dysregulation in both diseases is the differential expression of complement receptors CR1, CR2, CR3, and CR4. The rs1143679 variant of ITGAM, encoding the R77H variant of CD11b (part of complement receptor 3; CR3), is among the strongest genetic susceptibility effects in human SLE PubMed; this variant impairs CR3-mediated phagocytosis, adhesion, and regulatory suppression of innate immune signaling, providing human genetic validation of CR3 as a mechanistically relevant target. CR2 (CD21) lowers the activation threshold of autoreactive B cells by co-stimulating the B cell receptor in the presence of C3d-opsonized autoantigens, directly amplifying the autoantibody production characteristic of both SS and SLE. C5aR1 is a potent anaphylatoxin receptor that increases vascular permeability, induces oxidative bursting and release of proinflammatory cytokines in myeloid cells, and exerts chemotactic effects on myeloid and lymphoid cells ScienceDirect; its dysregulation drives NET formation and perpetuates the autoantigen burden in SLE. Collectively, the dysregulation of these receptors sustains a self-amplifying pro-inflammatory milieu that current therapies are insufficiently equipped to interrupt.

Despite the well-established pathogenic role of complement in SS and SLE, no approved therapy directly targets complement receptors in either disease. The C5a receptor inhibitor avacopan has demonstrated superior sustained remission at 52 weeks compared to standard prednisone taper in ANCA-associated vasculitis New England Journal of Medicine, providing proof-of-concept clinical validation for C5aR1 blockade in complement-mediated disease; however, this agent remains inadequately studied in SS and SLE. Notably, avacopan is currently under regulatory scrutiny due to data integrity concerns in the pivotal ADVOCATE trial The Rheumatologist, underscoring the need for robust mechanistic evidence to guide the development of complement receptor-directed therapies. The transcriptional landscape of complement receptor expression in SS and SLE has not been comprehensively characterized, limiting the ability to identify the most actionable therapeutic targets. Using publicly available RNA-seq datasets sourced from NCBI Gene Expression Omnibus (GEO), this study performs differential gene expression and pathway enrichment analyses comparing SS and SLE patients to healthy controls, with the aim of characterizing complement receptor dysregulation at the transcriptional level and providing molecular rationale for targeted complement receptor blockade as a therapeutic strategy in SS and SLE.

Results

Just for kicks here's a citation (10). And a reference to a supplement Sup. Note 1. And TODO List.
Figure **??** shows an example of how to insert a column-wide figure. To insert a figure wider than one column, please use the `\begin{figure*}...\end{figure*}` environment. Figures wider than one column should be sized to 11.4 cm or 17.8 cm wide. Use `\begin{SCfigure*}...\end{SCfigure*}` for a wide figure with side captions.

C3aR and C5aR Expression in Normal Brain and Kidney.

Enhanced Expression in Sjögren's and Lupus Tissues.

Pathway Enrichment.

Discussion

Our findings demonstrate that complement anaphylatoxin receptors are persistently upregulated in both central and peripheral target organs in chronic autoimmune disease such as sjogrens. This pattern underscores the dual nature of complement, vital for homeostasis but deleterious when uncontrolled. The strong expression in microglia parallels reports implicating complement in synaptic pruning and neurodegeneration, while renal upregulation aligns with complement driven glomerular and tubulointerstitial injury. Therapeutic inhibition of C3aR/C5aR signaling has shown benefit in preclinical models, and our transcriptomic results lend human relevance to these approaches. Emerging C3aR antagonists and C5aR inhibitors (e.g., avacopan) may thus hold promise for neuro-renal protection in Sjögren's. Integrating receptor profiling with spatial transcriptomics and proteomic validation will refine patient stratification for complement-targeted therapy.

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Conclusions

Complement receptors C3aR and C5aR are upregulated in autoimmune brain and kidney, bridging innate immunity and chronic inflammation. Their distribution across immune and parenchymal compartments identifies them as rational targets for tissue-directed complement modulation.

ACKNOWLEDGEMENTS
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COMPETING FINANCIAL INTERESTS
None.

Bibliography

1. Carlos Andrés Reyes Jaraba, Nehomar Eduardo Pájaro Galvis, Oscar Vicente Vergara Serpa, Johan Yessid Conquett Huertas, Oscar Darío Echenique Torres, Alonso Cortina Gutiérrez, Diego Antonio Serna Otero, and Diana Carolina Zapata Cerpa. Sjögren's syndrome: Epidemiology and clinical manifestations. Revista Colombiana de Reumatología (English Edition), 29(4):310–324, October 2022. ISSN 2444-4405. doi: 10.1016/j.rcreue.2020.12.006.
2. Eleanor Thurtle, Alice Grosjean, Monia Steenackers, Katharina Strege, Giovanna Barcelos, and Pushpendra Goswami. Epidemiology of sjögren's: A systematic literature review. Rheumatology and Therapy, 11(1):1–17, Feb 2024. ISSN 2198-6584. doi: 10.1007/s40744-023-00611-8.
3. Hong Huang, Wenhui Xie, Yan Geng, Yong Fan, and Zhuoli Zhang. Mortality in patients with primary sjä¶grenâ€™s syndrome: a systematic review and meta-analysis. Rheumatology, 60(9):4029–4038, 04 2021. ISSN 1462-0324. doi: 10.1093/rheumatology/keab364.
4. Jingru Tian, Dingyao Zhang, Xu Yao, Yaqing Huang, and Qianjin Lu. Global epidemiology of systemic lupus erythematosus: a comprehensive systematic analysis and modelling study. Annals of the Rheumatic Diseases, 82(3):351–356, March 2023. ISSN 0003-4967. doi: 10.1136/ard-2022-223035.
5. Megan R. W. Barber, Cristina Drenkard, Titilola Falasinnu, Alberta Hoi, Anselm Mak, Nien Yee Kow, Elisabet Svenungsson, Jonna Peterson, Ann E. Clarke, and Rosalind Ramsey-Goldman. Global epidemiology of systemic lupus erythematosus. Nature Reviews Rheumatology, 17(9):515–532, August 2021. ISSN 1759-4804. doi: 10.1038/s41584-021-00668-1.
6. Samir V. Parikh, Salem Almaani, Sergey Brodsky, and Brad H. Rovin. Update on lupus nephritis: Core curriculum 2020. American Journal of Kidney Diseases, 76(2):265–281, August 2020. ISSN 0272-6386. doi: 10.1053/j.ajkd.2019.10.017.
7. +Guillermo Ruacho, +Marika Kvarnström, +Agneta Zickert, +Vilija Oke, +Johan Rönnelid, +Susanna Eketjäll, +Kerstin Elvin, +Iva Gunnarsson, and +Elisabet Svenungsson. Sjögren syndrome in systemic lupus erythematosus: A subset characterized by a systemic inflammatory state. The Journal of Rheumatology, 47(6):865–875, 2020.
8. Catharina Eriksson, Heidi Kokkonen, Martin Johansson, Göran Hallmans, Göran Wadell, and Solbritt Rantapää-Dahlqvist. Autoantibodies predate the onset of systemic lupus erythematosus in northern sweden. Arthritis Research amp; Therapy, 13(1), February 2011. ISSN 1478-6362. doi: 10.1186/ar3258.
9. Marta Legatowicz-Koprowska, Stanisław Nitek, and Jolanta Czerwińska. The complement system in primary sjögren's syndrome: the expression of certain cascade and regulatory proteins in labial salivary glands – observational study. Rheumatology, 58(6):357–366, December 2020. ISSN 2084-9834. doi: 10.5114/reum.2020.102000.
10. Julien Carvelli, Olivier Demaria, Frédéric Vély, Luciana Batista, Nassima Chouaki Benmansour, Joanna Fares, Sabrina Carpentier, Marie-Laure Thibult, Ariane Morel, Romain Remark, Pascale André, Agnès Represa, Christelle Piperoglou, Laura Assante Miranda, William Baron, Nourhène Belaid, Clarisse Caillet, Flavien Caraguel, Barbara Carrette, Florent Carrette, Fabien Chanuc, Rachel Courtois, Aurore Fenis, Marilyn Giordano, Mathilde Girard-Madoux, Marc Giraudon-Paoli, Nicolas Gourdin, Gwendoline Grondin, Franceline Guillot, Guillaume Habif, Solène Jaubert, Julie Lopez, Mélanie Le Van, Naouel Lovera, Marine Mansuy, Elodie Bonnet, Audrey Sansaloni, Annick Reboul, Emmanuel Mitry, Camille Nekkar-Constant, Valentine Péri, Paul Ricaut, Léa Simon, Jean-Baptiste Vallier, Marie Vétizou, Robert Zerbib, Sophie Ugolini, Marion Etiennot, Justine Galluso, Luc Lyonnet, Jean-Marie Forel, Laurent Papazian, Lionel Velly, Baptiste André, Antoine Briantais, Benoit Faucher, Estelle Jean, Julie Seguier, Veronique Veit, Jean-Robert Harlé, Boris Pastorino, Clémence Delteil, Laurent Daniel, Jean-Paul Boudsocq, Axelle Clerc, Emmanuel Delmond, Pierre-Olivier Vidal, Hélène Savini, Bruno Coutard, Pierre Yves Cordier, Erwan Le Dault, Christophe Guervilly, Pierre Simeone, Marc Gainnier, Yannis Morel, Mikael Ebbo, Nicolas Schleinitz, Eric Vivier, the Explore COVID-19 IPH group, and the Explore COVID-19 Marseille Immunopole group. Association of covid-19 inflammation with activation of the c5a–c5ar1 axis. Nature, 588(7836):146–150, 2020. doi: 10.1038/s41586-020-2600-6.
11. Meiling Zheng, Zhi Hu, Xiaole Mei, Lianlian Ouyang, Yang Song, Wenhui Zhou, Yi Kong, Ruifang Wu, Shijia Rao, Hai Long, Wei Shi, Hui Jing, Shuang Lu, Haijing Wu, Sujie Jia, Qianjin Lu, and Ming Zhao. Single-cell sequencing shows cellular heterogeneity of cutaneous lesions in lupus erythematosus. Nature Communications, 13(1), December 2022. ISSN 2041-1723. doi: 10.1038/s41467-022-35209-1.
12. Richard K. Perez, M. Grace Gordon, Meena Subramaniam, Min Cheol Kim, George C. Hartoularos, Sasha Targ, Yang Sun, Anton Ogorodnikov, Raymund Bueno, Andrew Lu, Mike Thompson, Nadav Rappoport, Andrew Dahl, Cristina M. Lanata, Mehrdad Matloubian, Lenka Maliskova, Serena S. Kwek, Tony Li, Michal Slyper, Julia Waldman, Danielle Dionne, Orit Rozenblatt-Rosen, Lawrence Fong, Maria Dall'Era, Brunilda Balliu, Aviv Regev, Jinoos Yazdany, Lindsey A. Criswell, Noah Zaitlen, and Chun Jimmie Ye. Single-cell rna-seq reveals cell type–specific molecular and genetic associations to lupus. Science, 376(6589): eabf1970, 2022. doi: 10.1126/science.abf1970.
13. Yosta Vegting and Marc Hilhorst. Scrna-seq of human kidney immune cells of patients with anca-associated glomerulonephritis, lupus nephritis against a healthy nephrectomy control. https://www.ebi.ac.uk/biostudies/arrayexpress/studies/E-MTAB-13596, 2024.
14. Arnon Arazi, Deepak A. Rao, Celine C. Berthier, Anne Davidson, Yanyan Liu, Paul J. Hoover, Adam Chicoine, Thomas M. Eisenhaure, A. Helena Jonsson, Shuqiang Li, David J. Lieb, Fan Zhang, Kamil Slowikowski, Edward P. Browne, Akiko Noma, Danielle Sutherby, Scott Steelman, Dawn E. Smilek, Patti Tosta, William Apruzzese, Elena Massarotti, Maria Dall'Era, Meyeon Park, Diane L. Kamen, Richard A. Furie, Fernanda Payan-Schober, William F. Pendergraft, Elizabeth A. McInnis, Jill P. Buyon, Michelle A. Petri, Chaim Putterman, Kenneth C. Kalunian, E. Steve Woodle, James A. Lederer, David A. Hildeman, Chad Nusbaum, Soumya Raychaudhuri, Matthias Kretzler, Jennifer H. Anolik, Michael B. Brenner, David Wofsy, Nir Hacohen, and Betty Diamond. The immune cell landscape in kidneys of patients with lupus nephritis. Nature Immunology, 20(7):902–914, June 2019. ISSN 1529-2916. doi: 10.1038/s41590-019-0398-x.
15. Evan Der, Hemant Suryawanshi, Pavel Morozov, Manjunath Kustagi, Beatrice Goilav, Saritha Ranabothu, Peter Izmirly, Robert Clancy, H. Michael Belmont, Mordecai Koenigsberg, Michele Mokrzycki, Helen Rominieki, Jay A. Graham, Juan P. Rocca, Nicole Bornkamp, Nicole Jordan, Emma Schulte, Ming Wu, James Pullman, Kamil Slowikowski, Soumya Raychaudhuri, Joel Guthridge, Judith James, Jill Buyon, Thomas Tuschl, and Chaim Putterman. Tubular cell and keratinocyte single-cell transcriptomics applied to lupus nephritis reveal type i ifn and fibrosis relevant pathways. Nature Immunology, 20(7): 915–927, May 2019. ISSN 1529-2916. doi: 10.1038/s41590-019-0386-1.
16. Thomas J.F. Pranzatelli, Paola Perez, Anson Ku, Bruno Matuck, Khoa Huynh, Shunsuke Sakai, Mehdi Abed, Shyh-Ing Jang, Eiko Yamada, Kalie Dominick, Zara Ahmed, Amanda J. Oliver, Rachael Bogle, Quinn T. Easter, Alan N. Baer, Eileen Pelayo, Zohreh Khavandgar, David E. Kleiner, M. Teresa Magone, Sarthak Gupta, Christopher Lessard, A. Darise Farris, Peter D. Burbelo, Daniel Martin, Robert J. Morell, Changyu Zheng, Nicholas Rachmaninoff, Jose Maldonado-Ortiz, Katarzyna M. Tyc, Xufeng Qu, Marit Aure, Mohammad H. Dezfulian, Ross Lake, Sarah A. Teichmann, Daniel L. Barber, Lam C. Tsoi, Adam G. Sowalsky, Jinze Liu, Johann Gudjonsson, Kevin M. Byrd, Philip L.F. Johnson, John A. Chiorini, and Blake M. Warner. Gzmk+cd8+ t cells target a specific acinar cell type in sjögren's disease. Annals of the Rheumatic Diseases, October 2025. ISSN 0003-4967. doi: 10.1016/j.ard.2025.08.014.
17. Xiaoping Hong, Shuhui Meng, Donge Tang, Tingting Wang, Liping Ding, Haiyan Yu, Heng Li, Dongzhou Liu, Yong Dai, and Min Yang. Single-cell rna sequencing reveals the expansion of cytotoxic cd4+ t lymphocytes and a landscape of immune cells in primary sjögren's syndrome. Frontiers in Immunology, 11, February 2021. ISSN 1664-3224. doi: 10.3389/fimmu.2020.594658.

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Word Counts

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Notes on Nature Methods Brief Communication.

- Abstract: 3 sentences, 70 words.
- Main text: 3 pages, 2 figures, 1000-1500 words, more figures possible if under 3 pages

Statistics on word count.

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Words in headers: 42
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  63+20+13 (4/1/0/0) Section: Results
  120+1+0 (1/0/0/0) Section: Discussion
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Supplementary Note 1: TODO List

1. Add references
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2. Write abstract
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Supplementary Note 2: Dataset Outline

A. Lupus erythematosus.

1. GSE179633 (11)
   Summary: Lupus, a server and complex autoimmune disease, is clinically divided into cutaneous lupus erythematosus (CLE) which featured in skin damage, and systemic lupus erythematosus (SLE) which characterized in systemic multi-organ damage. The distinction of these two types of lupus is widely unknown. Here, we collected 23 skin biopsies of healthy control(HC), DLE (discoid lupus erythematosus, a main type of CLE) and SLE, separated epidermis and dermis and performed single cell RNA sequencing through microfluidics based 10x genomics system. Our results demonstrated larger numbers of immune cells infiltrated in skin lesions of DLE than SLE, which may help to distinguish them. Then, non-immune cells such as keratinocytes and fibroblasts were showed functions like immune cells. Moreover, ISGs(interferon stimulated genes), HSP70 coding genes were found to be overexpressed in multi expanded subclusters. Some biological progresses such as autophagy and neutrophil activation were enriched in expanded subclusters.
   Overall Design: We collected totally 23 skin biopsies from 5 HC, 8 DLE patients and 10 SLE patients and separated epidermis and dermis. Finally, 14 epidermal single-cell suspensions (4 HC,5 DLE and 5 SLE) and 16 dermal single-cell suspensions (4 HC,5 DLE and 7 SLE) were successfully prepared for scRNA-seq by 10× genomics.

2. GSE137029 (12)
   Description: Systemic lupus erythematosus (SLE) is a heterogeneous autoimmune disease. Knowledge of circulating immune cell types and states associated with SLE remains incomplete. We profiled over 1.2 million PBMCs (162 cases, 99 controls) with multiplexed single-cell RNA-sequencing (mux-seq). Cases exhibited elevated expression of type-1 interferon-stimulated genes (ISG) in monocytes, reduction of naïve CD4+ T cells that correlated with monocyte ISG expression, and expansion of repertoire-restricted cytotoxic GZMH+ CD8+ T cells. Cell-type-specific expression features predicted case-control status and stratified patients into two molecular subtypes. We integrated dense genotyping data to map cell-type-specific cis-eQTLs and link SLE-associated variants to cell-type-specific expression. These results demonstrate mux-seq as a systematic approach to characterize cellular composition, identify transcriptional signatures, and annotate genetic variants associated with SLE.

3. E-MTAB-13596 (13)
   Description: ANCA-associated glomerulonephritis (AGN) associates with a high risk of end-stage kidneydisease. The role of kidney immune cells in local inflammation remains unclear. Herewe investigate kidney immune cell diversity and function. Kidney tissue from AGN patients (n=5) and a lupus nephritis (LN) patient (n=1) were aquired during a biopsy procedure for a clinical indication. Needle-core biopsies were obtained for histopathological examination, and an additional pass was performed to retrieve kidney tissue for scRNA-seq. Healthy kidney tissue (n=1) was obtained from a kidney that was surgically removed do tue due to a (non-invasive) papillary urothelial carcinoma. Immediately after collection, kidney tissue was processed into a single-cell suspension and sorted using a 4-color flow cytometry panel to isolate living, CD45+immune cells. To aid in the multi-omic characterization, surface markers and T and B cell repertoires were sequenced in 2 samples (1 AGN patient and the nephrectomy control). These samples were incubated with an oligo-antibody TotalSeq-C cocktail containing 130 unique cell surface antigens.

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4. AMP Phase 1 (14) (15)
   Description (Broad): Lupus nephritis is a potentially fatal autoimmune disease, whose current treatment is ineffective and often toxic. To gain insights into disease mechanisms, we analyzed kidney samples from lupus nephritis patients and healthy controls using single-cell RNA-seq. Our analysis revealed 21 subsets of leukocytes active in disease, including multiple populations of myeloid, T, NK and B cells, demonstrating both pro-inflammatory and resolving responses. We found evidence of local activation of B cells correlated with an age-associated B cell signature, and of progressive stages of monocyte differentiation within the kidney. A clear interferon response was observed in most cells. Two chemokine receptors, CXCR4 and CX3CR1, were broadly expressed, pointing to potential therapeutic targets. Gene expression of immune cells in urine and kidney was highly correlated, suggesting urine may be a surrogate for kidney biopsies. Our results provide a first comprehensive view of the complex network of leukocytes active in lupus nephritis kidneys.
   Description (Metro): Lupus nephritis (LN) occurs in up to 50% of patients with systemic lupus erythematosus (SLE), and is a major contributor to mortality and morbidity. LN presents as a highly heterogeneous disease both in histopathology and response to therapy. The molecular and cellular processes leading to renal damage and to the heterogeneity of the disease are not well understood. To elucidate the processes underpinning the heterogeneity of LN, we applied single cell RNA-sequencing (scRNA-seq) to renal biopsies from LN patients. Skin biopsies were evaluated as a source of biomarkers for monitoring kidney disease. Type-I interferon (IFN) response signatures were identified in tubular cells and keratinocytes, differentiating LN patients from healthy controls. Non-responders associated with higher IFN signatures in both tissue compartments. Moreover, non-response was also associated with a fibrotic signature in the tubular cells. Receptor-ligand interaction analysis indicated that the fibrotic process is likely mediated by FGF receptors with the initiating signal originating from infiltrating leukocytes. Differential expression analysis of tubular cells between proliferative and membranous LN pointed to several fibrosis-relevant pathways, which may offer insight into their histological differences. In summary, scRNA-seq was applied to LN to deconstruct its heterogeneity and provide novel targets for personalized approaches to therapy.

B. Sjögren's syndrome.

1. CellXGene (16)
   Description: Sjögren's Disease (SjD) is a systemic autoimmune disease without a clear etiology or effective therapy. Utilizing unbiased single-cell and spatial transcriptomics to analyze human minor salivary glands in health and disease we developed a comprehensive understanding of the cellular landscape of healthy salivary glands and how that landscape changes in SjD patients. This study explores the complex interplay of varied cell types in the salivary glands and their role in the pathology of Sjögren's Disease.

2. GSE157278 (17)
   Summary: By single cell RNA sequencing,our data revealed disease-specific immune cell subsets and provide some potential new targets of pSS, specific expansion of CD4+ CTLs may be involved in the pathogenesis of pSS, which might give a valuable insights for therapeutic interventions of pSS.
   Overall Design: We applied single cell RNA sequencing (scRNA-seq) to 57, 288 peripheral blood mononuclear cells (PBMCs) from 5 patients with pSS and 5 healthy controls. The immune cell subsets and susceptibility genes involved in the pathogenesis of pSS were analyzed.

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